

Ph.D. Exam in Analysis May 2012

Be sure to carefully present all work in a neat and logical fashion. Do not leave any gaps. State clearly theorems used in your proofs. Print your name on each sheet turned in.

- (1) Let μ be a real-valued signed measure on a σ -algebra Σ . Does there exist a set in Σ on which μ attains its minimum value? Prove or disprove.
- (2) Let X be a Banach space. Show that there exists a compact T_2 space S such that X is isometrically isomorphic to a closed linear subspace of $C(S)$.
[Hint: Examine the unit ball of the dual space X^* with the appropriate topology and the action of X on X^*]
- (3) Let $\{f_n\}$ be a sequence of integrable functions on S such that $\sum \int |f_n| d\mu < \infty$. Give all details concerning the convergence of $\sum f_n(s)$, $s \in S$.
- (4) State and prove the Fubini theorem.
- (5) Let π be the natural map from the Banach space X into its second dual X^{**} . Show $\pi(X)$ is closed in the norm topology of X^{**} .
- (6) Show that the continuous functions on $[0, 1]$ are dense in $L^1[0, 1]$ (Lebesgue measure).
[Hint: First show this is true for the indicator function of a Lebesgue measurable subset of $[0, 1]$]